

Set Up Your Environment

In this guide, you'll learn how to set up your environment, so that you can run your project with Android Studio and Xcode. This will allow you to develop with Android emulators and iOS simulators, build your app locally, and more.

NOTE

This guide requires Android Studio or Xcode. If you already have one of these programs installed, you should be able to get up and running within a few minutes. If they are not installed, you should expect to spend about an hour installing and configuring them.

▶ Is setting up my environment required?

Development OS

macOS

Windows

Linux

Target OS

Android

iOS

Installing dependencies

You will need Node, the React Native command line interface, a JDK, and Android Studio.

While you can use any editor of your choice to develop your app, you will need to install Android Studio in order to set up the necessary tooling to build your React Native app for Android.

Node

Follow the [installation instructions for your Linux distribution](#) to install Node 18.18 or newer.

Java Development Kit

React Native currently recommends version 17 of the Java SE Development Kit (JDK). You may encounter problems using higher JDK versions. You may download and install [OpenJDK](#) from [AdoptOpenJDK](#) or your system packager.

Android development environment

Setting up your development environment can be somewhat tedious if you're new to Android development. If you're already familiar with Android development, there are a few things you may need to configure. In either case, please make sure to carefully follow the next few steps.

1. Install Android Studio

[Download and install Android Studio](#). While on Android Studio installation wizard, make sure the boxes next to all of the following items are checked:

- Android SDK
- Android SDK Platform
- Android Virtual Device

Then, click "Next" to install all of these components.

If the checkboxes are grayed out, you will have a chance to install these components later on.

Once setup has finalized and you're presented with the Welcome screen, proceed to the next step.

2. Install the Android SDK

Android Studio installs the latest Android SDK by default. Building a React Native app with native code, however, requires the Android 15 (VanillaIceCream) SDK in particular. Additional Android SDKs can be installed through the SDK Manager in Android Studio.

To do that, open Android Studio, click on "Configure" button and select "SDK Manager".

The SDK Manager can also be found within the Android Studio "Settings" dialog, under **Languages & Frameworks → Android SDK**.

Select the "SDK Platforms" tab from within the SDK Manager, then check the box next to "Show Package Details" in the bottom right corner. Look for and expand the Android 15 (VanillaIceCream) entry, then make sure the following items are checked:

- Android SDK Platform 35
- Intel x86 Atom_64 System Image or Google APIs Intel x86 Atom System Image

Next, select the "SDK Tools" tab and check the box next to "Show Package Details" here as well. Look for and expand the "Android SDK Build-Tools" entry, then make sure that 35.0.0 is selected.

Finally, click "Apply" to download and install the Android SDK and related build tools.

3. Configure the **ANDROID_HOME** environment variable

The React Native tools require some environment variables to be set up in order to build apps with native code.

Add the following lines to your `$HOME/.bash_profile` or `$HOME/.bashrc` (if you are using `zsh` then `~/.zprofile` or `~/.zshrc`) config file:

shell

```
export ANDROID_HOME=$HOME/Android/Sdk
export PATH=$PATH:$ANDROID_HOME/emulator
export PATH=$PATH:$ANDROID_HOME/platform-tools
```

`.bash_profile` is specific to `bash`. If you're using another shell, you will need to edit the appropriate shell-specific config file.

Type `source $HOME/.bash_profile` for `bash` or `source $HOME/.zprofile` to load the config into your current shell. Verify that `ANDROID_HOME` has been set by running `echo $ANDROID_HOME` and the appropriate directories have been added to your path by running `echo $PATH`.

Please make sure you use the correct Android SDK path. You can find the actual location of the SDK in the Android Studio "Settings" dialog, under **Languages & Frameworks** → **Android SDK**.

Watchman

Follow the [Watchman installation guide](#) to compile and install Watchman from source.

[Watchman](#) is a tool by Facebook for watching changes in the filesystem. It is highly recommended you install it for better performance and increased compatibility in certain edge cases (translation: you may be able to get by without installing this, but your mileage may vary; installing this now may save you from a headache later).

Preparing the Android device

You will need an Android device to run your React Native Android app. This can be either a physical Android device, or more commonly, you can use an Android Virtual Device which allows you to emulate an Android device on your computer.

Either way, you will need to prepare the device to run Android apps for development.

Using a physical device

If you have a physical Android device, you can use it for development in place of an AVD by plugging it in to your computer using a USB cable and following the instructions [here](#).

Using a virtual device

If you use Android Studio to open `./AwesomeProject/android`, you can see the list of available Android Virtual Devices (AVDs) by opening the "AVD Manager" from within Android Studio. Look for an icon that looks like this:



If you have recently installed Android Studio, you will likely need to [create a new AVD](#). Select "Create Virtual Device...", then pick any Phone from the list and click "Next",

then select the **VanillaIceCream** API Level 35 image.

We recommend configuring [VM acceleration](#) on your system to improve performance. Once you've followed those instructions, go back to the AVD Manager.

Click "Next" then "Finish" to create your AVD. At this point you should be able to click on the green triangle button next to your AVD to launch it.

That's it!

Congratulations! You successfully set up your development environment.



Now what?

- If you want to add this new React Native code to an existing application, check out the [Integration guide](#).
- If you're curious to learn more about React Native, check out the [Introduction to React Native](#).

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